

Worksheet 11 – Understanding cophenetic distances

- (a) In the previous exercise (Worksheet 10), you used Ward's method to cluster the 4 items below whose measurements on a single variable X are given in the table:

Item	Measurement x
1	7
2	10
3	22
4	24

Calculate the cophenetic distances by hand for every pair of items.

- (b) Calculate the correlation between the cophenetic distance and the original distances by hand.

$$r = \frac{1}{n-1} \sum \left(\frac{x - \bar{x}}{s_x} \right) \left(\frac{y - \bar{y}}{s_y} \right)$$

Update: It is too cumbersome to calculate correlation by hand (although it is a good exercise if you have the time). For this part (b), use R to obtain the correlation between cophenetic distance calculated in part (a) and the original distance.

- (c) Write R code using `cophenetic()` to validate your answers in parts (a) and (b).